Prone Positioning

Guideline Title: Prone Positioning	Guideline Number: GUID-3203-PR029

Issue date: 1/8/2021	Replaces Dept. Policy:
Revision dates: 6/8/2021	Developed by: Air Care Team
Approved by: Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature:	Signature:
Department Numbers: 3203	

CCCLII. **PURPOSE:** Prone positioning of patients with Acute Respiratory Distress Syndrome (ARDS) may lead to improved oxygenation and decreased mortality. The mechanism of benefit from prone positioning is multifactorial, including improved lung recruitment, improved oxygenation, and reduced pleural pressure gradient across the lungs, leading to more effective distribution of tidal volume.

CCCLIII. **DEFINITIONS:**

A. ACT – Air Care Team
B. ARDS – Acute Respiratory Distress Syndrome
C. FPM – Flight paramedic
D. FRN – Flight registered nurse

CCCLIV. **DEPARTMENT GUIDELINE:**

A. Prone positioning of mechanically ventilated patient initiated by sending physician
    i. ASSESSMENT/INCLUSION CRITERIA
        1. Severe ARDS
        2. Other pulmonary conditions believed to be improved by repositioning
    ii. ASSESSMENT/EXCLUSION CRITERIA
        1. Inability to tolerate positioning
B. Many prone patients are unstable and have several lines, tubes, and monitors, take care to be deliberate in movement and transfer
C. **Prepare patient for transfer**
    i. Flight nurse leads crew with “time out” and brief of intended course of action
    ii. Connect patient to ACT ventilator per *GUID-PR024 - Ventilation with a Mechanical Ventilator*.
        1. Use most appropriate transport settings to maintain current clinical needs
        2. Patient should be on ACT ventilator minimum of 10 minutes

prior to transfer to assess for negative changes

iii. Transition patient to ACT monitor. Place ECG electrodes to posterior thorax.
iv. Drape lines up and over right shoulder
v. Place defibrillator pads in the best possible position; anterior-posterior or anterior-lateral position. Posterior-lateral and Bi-axillary can be acceptable
vi. Ensure all lines and tubes are secure
vii. Consider analgesia & sedation management per *GUID-3203-G005 Analgesia and Sedation Management or GUID-G007 - Propofol Diprivan* (if ordered by sending facility) *to maintain RASS scale -4.*
    1. Consider Rocuronium (Zemuron) 0.6-1.2mg/kg IVP/IO (adult) or 0.6mg/kg IVP (peds) for improved ventilator synchronicity during transport if patient is intubated/mechanically ventilated and has adequate sedation. **Do not give a paralytic drug if patient cannot be sedated**
viii. Place patient's arms at their side (neutral position)
ix. Position the transport stretcher next to the hospital bed
x. Place padding parallel to the patient's chest and hips on stretcher, if not already underneath and coming with the patient
xi. Have intubation and airway management equipment at bedside and prepared prior to moving patient
xii. Have a minimum of 5 staff at bedside to safely move patient

D. Transfer Patient
    i. ACT FPM at head of bed (HOB), controlling movement and securing ETT
        1. ACT FPM at HOB will be intubator, if needed
    ii. Use transfer sheet or board for movement
    iii. Any participant will call "STOP" during the procedure if concerned
    iv. ACT FRN + hospital staff next to our stretcher
    v. 2 hospital staff next to hospital bed
    vi. Patient movement coordinated from provider at HOB
    vii. Move patient in slow, controlled stages across bed to stretcher
    viii. Be mindful of endotracheal tube (ETT), circuit, IV lines, and tubes during move
    ix. Move small amount, reposition lines / tubes as needed; repeat
    x. Patient's face is toward right shoulder, with ETT/circuit off right shoulder

E. Post-transfer Assessment
    i. Reassess patient (Vital signs/hemodynamic /dysrhythmias, ETCO2, worsening gas exchange, lines/tubes,)
    ii. Observe patient for clinical change for minimum 10 minutes after moving
    iii. Check patient placement & padding: Eyes closed, ears not folded, head not hyperextended, no pressure on eyes/nose. Pad face as

needed. (Reassess en route)

iv. Assure sedation, analgesia, and paralysis is uninterrupted en route

F. Resuscitation and Cardiac Arrest

i. Turning a critically ill prone patient in an emergency is associated with significant risk. There is risk of displacement of the endotracheal tube, removal of lines or tubes, as well as injury to the patient and staff. The time delay associated with the procedure would also inevitably delay effective chest compressions and defibrillation. *Literature has shown posterior compressions to be at least as effective as supine compressions if performed correctly*.

ii. If the prone positioned patient suffers cardiac arrest during transport, resuscitation should be attempted in the prone position, with posterior compressions; otherwise following normal ACLS guidelines appropriate for the scenario.

1. compressions:

a. The heel of the compressor's hand should be placed in the center of the upper back, at the level of the 7th thoracic vertebrae. (Between scapulae)

G. Notify receiving facility of positioning to ensure adequate resources will be available to transfer patient upon arrival

CCCLV. **DOCUMENTATION:**

CCCLVI. **REFERENCES:**

A. Drahnak, D. M. & Custer, N. (2015). Prone positioning of patients with acute respiratory distress syndrome. *CriticalCareNurse, 35*(6), 29-37.

B. Bell, L. (2020). ARDS, COVID-19 and Pronation Therapy. *American Association of Critical-Care Nurses*. https://www.aacn.org/blog/ards-covid-19-and-pronation-therapy